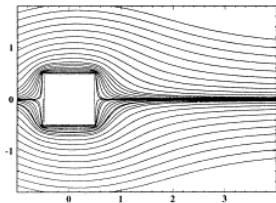


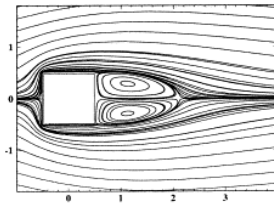
Equations to be solved + Large Eddy Simulation

$$\nabla \cdot \mathbf{u} = 0$$

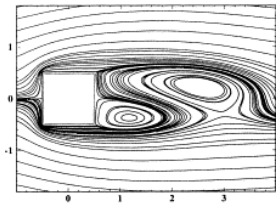
$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u}$$



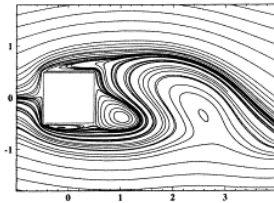
(a)



(b)



(c)



(d)

LBM algorithm

- Initialize DFs e.g. $f_i^{\text{eq}}(\rho_{\text{ini}}, \mathbf{u}_{\text{ini}})$
- Setup τ according to material properties
- Repeat until steady state or desired time is achieved
 1. Compute macroscopic quantities from f_i
 2. Compute equilibrium DF's and post-collision state
 3. Stream f_i^* along lattice links
 4. Apply BC

$$f_i(t+1, \mathbf{x} + \mathbf{c}_i) = f_i(t, \mathbf{x}) + \mathcal{M}^{-1} \mathcal{S}(m_i^{\text{eq}}(t, \mathbf{x}) - m_i(t, \mathbf{x})) = f_i^*$$

$$m_i^{\text{eq}} = \mathcal{M} f_i^{\text{eq}} \quad m_i = \mathcal{M} f_i$$

$$\nu_{lb} = c_s^2 \left(\tau - \frac{1}{2} \right) + \nu_t,$$

$$f_i^{\text{eq}}(t, \mathbf{x}) = w_i \rho(t, \mathbf{x}) \left(1 + \frac{\mathbf{c}_i \cdot \mathbf{u}(t, \mathbf{x})}{c_s^2} + \frac{(\mathbf{c}_i \cdot \mathbf{u})^2}{2c_s^4} - \frac{\mathbf{u}^2}{2c_s^2} \right)$$

macro-variables...

$$\rho = \sum_i f_i(t, \mathbf{x}) \quad \rho \mathbf{u} = \sum_i \mathbf{c}_i f_i(t, \mathbf{x})$$

Multiple Relaxation Time LBM

Stability of SRT can be improved by performing collision in moment space,

$$f_i(x + c_i \delta t, t + \delta t) - f_i(x, t) = -\mathbb{M}^{-1} \mathbb{S} [\mathbf{m}(x, t) - \mathbf{m}^{eq}(x, t)]$$

$\mathbf{m} = (m_0, m_1, \dots, m_n)^T$ are vectors of moments, $\mathbb{S}$ is diagonal matrix of relaxation times for each moment and mapping between distribution and moment spaces is given by

$$\mathbf{m} = \mathbb{M} \mathbf{f}, \quad \mathbf{f} = \mathbb{M}^{-1} \mathbf{m}$$

for D2Q9 model the vector of moments reads

$$\mathbf{m} = (\rho, e, \epsilon, j_x, q_x, j_y, q_y, p_{xx}, p_{xy})^T$$

and the equilibrium moment vector reads

$$\mathbf{m}^{eq} = (\rho, -2\rho + 3(j_x^2 + j_y^2), \rho - 3(j_x^2 + j_y^2), j_x, -j_x, j_y, -j_y, (j_x^2 - j_y^2), j_x j_y)^T$$

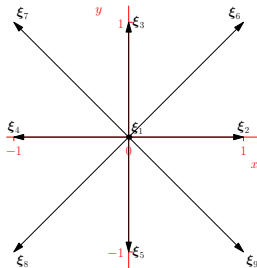
MRT Matrix

	C	E	N	W	S	NE	NW	SW	SE
m_ρ	1	1	1	1	1	1	1	1	1
m_e	-4	-1	-1	-1	-1	2	2	2	2
m_ϵ	4	-2	-2	-2	-2	1	1	1	1
m_{j_x}	0	1	0	-1	0	1	-1	-1	1
m_{q_x}	0	-2	0	2	0	1	-1	-1	1
m_{j_y}	0	0	1	0	-1	1	1	-1	-1
m_{q_y}	0	0	-2	0	2	1	1	-1	-1
$m_{p_{xx}}$	0	1	-1	1	-1	0	0	0	0
$m_{p_{xy}}$	0	0	0	0	0	1	-1	1	-1

MRT Inverse Matrix

	m_ρ	m_e	m_ϵ	m_{j_x}	m_{q_x}	m_{j_y}	m_{q_y}	$m_{p_{xx}}$	$m_{p_{xy}}$
f_C	1/9	-1/9	1/9	0	0	0	0	0	0
f_E	1/9	-1/36	-1/18	1/6	-1/6	0	0	1/4	0
f_N	1/9	-1/36	-1/18	0	0	1/6	-1/6	-1/4	0
f_W	1/9	-1/36	-1/18	-1/6	1/6	0	0	1/4	0
f_S	1/9	-1/36	-1/18	0	0	-1/6	1/6	-1/4	0
f_{NE}	1/9	1/18	1/36	1/6	1/12	1/6	1/12	0	1/4
f_{NW}	1/9	1/18	1/36	-1/6	-1/12	1/6	1/12	0	-1/4
f_{SW}	1/9	1/18	1/36	-1/6	-1/12	-1/6	-1/12	0	1/4
f_{SE}	1/9	1/18	1/36	1/6	1/12	-1/6	-1/12	0	-1/4

LBM D2Q9 Model



	C	E	N	W	S	NE	NW	SW	SE
c_i	(0,0)	(1,0)	(0,1)	(-1,0)	(0,-1)	(1,1)	(-1,1)	(-1,-1)	(1,-1)
t_i	4/9	1/9	1/9	1/9	1/9	1/36	1/36	1/36	1/36

lattice speed of sound $c_s = \frac{1}{\sqrt{3}}$

Large Eddy Turbulence Model

Extra viscosity term from the turbulence model (eddy viscosity):

$$\nu_t = (C\Delta)^2 |\mathbb{S}| = (C\Delta)^2 \sqrt{2S_{ij}S_{ij}}$$

Strain-rate tensor and it's contraction:

$$S_{ij}S_{ij} = (S_{xx}^2 + 2S_{xy}^2 + S_{yy}^2) = \left(\frac{\partial u_x}{\partial x}\right)^2 + \frac{1}{2} \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right)^2 + \left(\frac{\partial u_y}{\partial y}\right)^2$$

Partial derivatives of u_x and u_y is needed (FD, etc), but we can obtain whole $S_{\alpha\beta}$ from non-equilibrium DF or moments

$$2S_{\alpha\beta} = \frac{m_{\alpha\beta}^{neq}}{-\rho\tau c_s^2}$$

where

$$m_{\alpha\beta}^{neq} = \sum_i c_{i\alpha} c_{i\beta} (f_i - f_i^{eq})$$

LES Implementation

- Compute $\Pi^{\text{neq}} = \Pi - \Pi^{\text{eq}}$, where $\Pi_{ij} = m_{ij}$
- Compute $\tau_t = .5 \left(\sqrt{\tau^2 + 18C^2\Delta_x^2|\mathbb{S}|} + \tau \right)$, where $\tau = 3\nu_{lb} + 0.5$
- Final form of relaxation time is $\tau = .5 \left(\sqrt{\tau^2 + C^2\sqrt{648Q}/\rho} + \tau \right)$

$$\mathbb{S}_{ij} = \frac{3}{2\rho\tau} m_{ij}^{\text{neq}} \quad m_{ij}^{\text{eq}} = \frac{\rho}{3} \delta_{ij} + \rho u_i u_j \quad |\mathbb{S}| = (\mathbb{S}_{ii}^2 + 2(\mathbb{S}_{xy}^2 + \mathbb{S}_{xz}^2 + \mathbb{S}_{yz}^2))^{\frac{1}{2}}$$

$$m_{xx} = \sum_i f_i c_{i,x}^2 \quad m_{yy} = \sum_i f_i c_{i,y}^2 \quad m_{xy} = \sum_i f_i c_{i,x} c_{i,y}$$

Where Q can be computed from moments

$$Q = (m_{xx} - m_{xx}^{\text{eq}})^2 + (m_{yy} - m_{yy}^{\text{eq}})^2 + 2(m_{xy} - m_{xy}^{\text{eq}})^2$$

m_{xx} and m_{yy} from $\mathbb{M}^{-1}$, $m_{xy} = m_{pxy}$

$$m_{xx} = \sum_{i \in \{E, W, NE, NW, SE, SW\}} f_i \quad m_{yy} = \sum_{i \in \{N, S, NE, NW, SE, SW\}} f_i$$

LES Implementation cont'd

$$m_{xx} = \frac{2}{3}\rho + \frac{1}{6}m_e + \frac{1}{2}m_{pxx} \quad m_{yy} = \frac{2}{3}\rho + \frac{1}{6}m_e - \frac{1}{2}m_{pxx}$$

$$m_{xx}^{eq} = \frac{1}{3}\rho + \rho u_x^2 \quad m_{yy}^{eq} = \frac{1}{3}\rho + \rho u_y^2 \quad m_{xy}^{eq} = \rho u_x u_y$$

$$m_{xx} - m_{xx}^{eq} = \frac{1}{3}\rho - \frac{m_{jx}^2}{\rho} + \frac{1}{6}m_e + \frac{1}{2}m_{pxx}$$

$$m_{yy} - m_{yy}^{eq} = \frac{1}{3}\rho - \frac{m_{jy}^2}{\rho} + \frac{1}{6}m_e - \frac{1}{2}m_{pxx}$$

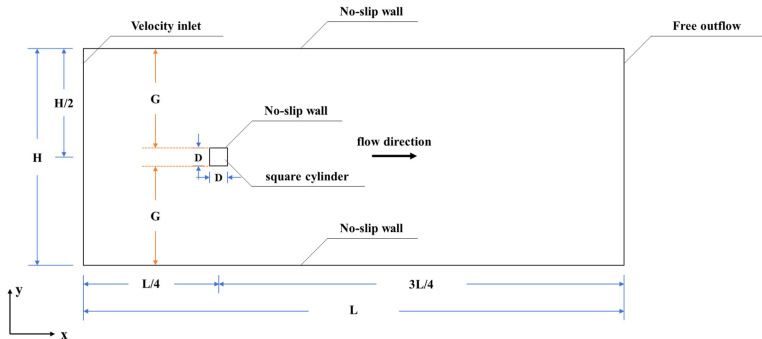
$$m_{xy} - m_{xy}^{eq} = m_{pxy} - \frac{m_{jx}m_{jy}}{\rho}$$

$$Q = \left(\frac{\rho}{3} - \frac{m_{jx}^2}{\rho} + \frac{m_e}{6} + \frac{m_{pxx}}{2} \right)^2 + \left(\frac{\rho}{3} - \frac{m_{jy}^2}{\rho} + \frac{m_e}{6} - \frac{m_{pxx}}{2} \right)^2 + 2 \left(m_{pxy} - \frac{m_{jx}m_{jy}}{\rho} \right)^2$$

Physical & lattice units relations

Reynolds and Mach numbers are used to setup the LBM parameters

$$u_{x,lb} = Ma\sqrt{c_s^2} \quad \nu_{lb} = \frac{u_{x,lb}L_{Y,lb}}{Re}$$



Initial and boundary conditions

Initial conditions:

$$\rho_{lb}(t = 0, \mathbf{x}) = 1 \quad \mathbf{u}_{lb}(t = 0, \mathbf{x}) = (0, 0)$$

Boundary conditions:

$$\rho_{lb,in} = 1 \quad \mathbf{u}_{lb,in} = Ma \cdot c_s$$

$$\mathbf{u}_{wall} = 0 \quad (\text{BounceBack halfway})$$

$$\rho_{out} = \rho(\mathbf{x}_{out} - dx\mathbf{n}_{out}) \quad \mathbf{u}_{out} = \mathbf{u}(\mathbf{x}_{out} - dx\mathbf{n}_{out})$$